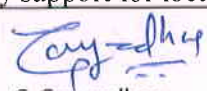
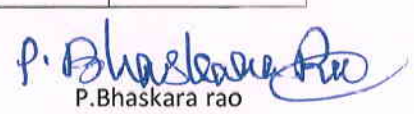


TECHNICAL SPECIFICATIONS FOR "SITC OF CCTV SYSTEM FOR ATC/TECHNICAL BUILDING & REMOTE SITES AT GVIAL AIRPORT,BHOGAPURAM

S.No	parameter	Minimum Requirement	Compliance (C/NC)
1	4MP IP Bullet Camera		
	Camera Type	IP Based Fixed Bullet type Camera (Pole & Wall mount compatible)	
	Image Device:	Bullet type camera; 1/2.8" Progressive scan CMOS or better	
	Lens :	OEM Motorized Varifocal lens, Auto Iris, lens f ~ 2.8 mm to 13.5 mm or better	
	Number of Pixels :	4MP or better	
	Resolution:	1080p HD or better	
	Minimum Illumination	Color mode: 0.005 lux @ F2.0 or better B/W: 0 lux with IR illumination on	
	Standard	CE or STQ Certified or BIS Standards	
	Compliance	Termination Box, Integrated RJ 45 and power connector provision should be provided with each camera	
	Compliance	ONVIF (Open Network Video Interface Forum) Profile 'S' & 'G' or better or interoperability compliance.	
	Video Stream:	H.265+, H.265, Dual H.264 or Better video stream for independent viewing and recording	
	Streams & fps	Triple streams ,individually Configurable; Main Stream:(1920x1080,1280x720,704x576) @25fps [PAL]; Sub Stream:(704x576,640x360)@25fps [PAL]; Third Stream: OEM to specify;	
	Electronic Shutter:	Auto	
	Auxiliary inputs & outputs:	1 alarm input, 1 relay output	
	Iris Control:	Auto	
	White balance	Auto	
	Wide Dynamic Range (WDR):	True WDR (120 dB or better)	
	Remote access:	Via browser for configuration, viewing and control	
	Ethernet:	10/100 Base-T Auto sensing, Half/ Full Duplex (RJ45)	
	SD Card:	One built-in Micro SD/SDHC/SDXC card slot of up to 128GB memory support for local storage (64 GB	

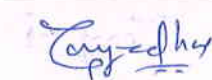

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		memory Card to be provided for local storage in case of network/Server failure)	
	Power Supply:	Power Should be drawn on PoE. Shall have ability to be powered by PoE+. IEEE 802.3 as well as normal DC/AC power.	
	Housing :	IP 66/67 or better, IK 10 certified for vandal proof	
	Mount:	Surface/Ceiling/pipe/corner/parapet/pendant with wall mount bracket/In Ceiling/ PVC Box as per site requirement. The camera mount should be of the same make as that of camera and suitable for the model number offered as specified by the manufacturer and should be an integrated unit. Should be compact and indoor/outdoor type as required Should support the weight of camera and accessories such as housing in any vertical or horizontal position etc	
	S/N Ratio	≥ 55dB for improved signal quality	
	IR Illuminator	Minimum of 50 mtrs built-in	
	Reset button	Shall be available	
	Operating Temperature	: 0 to +55 degree or better	
	Camera Boxes	PVC Camera boxes of Suitable sizes	
2	4MP IP Dome Camera		
	Camera Type	IP Based Dome Camera (/Ceiling/Wall mount compatible)	
	Image Device:	1/2.8" Progressive scan CMOS or better	
	Lens :	Auto Iris,	
	Number of Pixels :	Minimum 4MP Full HD or better.	
	Resolution:	1080p HD or better	
	Minimum Illumination	Color mode: 0.005 lux @ F2.0 or better B/W: 0 lux with IR illumination on	
	Standard	IR Cut Filter for Day/Night switching	
	Compliance	CE or STQ Certified or BIS Standards	
	Compliance	Termination Box, Integrated RJ 45 and power connector should be provided with each camera	
	Compliance	ONVIF (Open Network Video Interface Forum) Profile 'S' & 'G' or interoperability compliance..	
	Video Stream:	H.265+, H.265, Dual H.264 or Better video stream for independent viewing and recording	
		Triple streams ,individually Configurable; Main Stream:(1920x1080,1280x720,704x576)	


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	Streams & fps	@25fps [PAL]; Sub Stream:(704×576,640×360)@25fps [PAL]; Third Stream: OEM to specify; All streams should support adjustable frame rates up to 30fps	
	Electronic Shutter:	Auto	
	Auxiliary inputs & outputs:	1 alarm input, 1 relay output	
	Iris Control:	Auto & Support for manual iris override if required	
	White balance	Auto	
	Wide Dynamic Range (WDR):	True WDR, minimum 120dB or better	
	Remote access:	Via browser for configuration, viewing and control	
	Ethernet:	10/100 Base-T Auto sensing, Half/ Full Duplex (RJ45)	
	SD Card:	One built-in Micro SD/SDHC/SDXC card slot of up to 128GB memory support for local storage (64 GB memory Card to be provided for local storage in case of network/Server failure)	
	Power Supply:	Power Should be drawn on PoE. Shall have ability to be powered by PoE+. IEEE 802.3 as well as normal DC/AC power.	
	Housing :	IP 66/67 or better shall be of same make as of camera	
	Mount:	Surface/Ceiling/pipe/corner/parapet/pendant with wall mount bracket/In Ceiling/ PVC Box as per site requirement. The camera mount should be of the same make as that of camera and suitable for the model number offered as specified by the manufacturer and should be an integrated unit. Should be compact and indoor/outdoor type as required Should support the weight of camera and accessories such as housing in any vertical or horizontal position etc	
	S/N Ratio	≥ 55dB for improved signal quality	
	IR Illuminator	Minimum of 30m built-in IR Illuminator	
	Surge protection	Internal/External (In case of External catalogue of surge protector shall be submitted)	
	Reset button	Shall be available	
	Operating Temperature	: 0 to +55 degree or better	
	Camera License	Camera should be supplied with a valid, perpetual license for all included features	
	Camera Boxes	PVC Camera boxes of Suitable sizes	

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3	4MP IP Outdoor Dome Type PTZ Camera	
	Camera Type	IP Based Outdoor Dome type PTZ Camera (Wall/Pendent/ceiling mount compatible)
	Optical zoom (For Outdoor Camera)	Minimum 25 x Optical Zoom with Autofocus
	Number of Pixels	4 MP or better
	Resolution	1080p HD @ 25 FPS or better
	Image Device	1/2.8" Progressive scan CMOS or better
	Min. Illumination	Color mode: 0.005 lux @ F2.0 or better B/W: 0 lux with IR illumination ON
	Wide Dynamic Range	True WDR (120 dB or better)
	Video compression	H.265+, H.265, Smart Codec Support
	Streams & fps	Triple streams ,individually configurable; Main Stream:(1920x1080,1280x720,704x576) @25fps [PAL]; Sub Stream:(704x576,640x360)@25fps [PAL]; Third Stream:(1920x1080,1280x720, 704x576,640x480) @25fps [PAL];
	Pan Travel	360° Continuous
	Tilt Travel	0° - 90°, Auto Flip
	Manual Tilt Speed	0.5°/SEC to 90°/SEC
	Manual Pan Speed	Up to 400°/SEC
	Preset Tilt Speed:	0.5°/SEC to 90°/SEC
	Preset Pan Speed:	Up to 400°/SEC
	Standard	CE or STQ Certified or BIS Standards
	Compliance	Termination Box, Integrated RJ 45 and power connector should be provided with each camera
	Compliance	ONVIF (Open Network Video Interface Forum) Profile 'S' & 'G' or interoperability compliance..
	Auxiliary inputs & outputs	1 alarm input, 1 relay output
	Privacy masking zones:	4 or more

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	Iris control	Auto	
	Focus:	Auto	
	White balance:	Auto	
	Electronic shutter:	Auto	
	Remote access:	Via browser for configuration, viewing and control	
	Ethernet :	10/100/1000 (Gigabit) Ethernet Auto sensing, Half/Full Duplex (RJ45)	
	S/N Ratio:	>= 50 dB	
	SD Card slot :	One built-in Micro SD/SDHC/SDXC card slot of up to 128GB memory support for local storage (64 GB memory Card to be provided for local storage in case of network/Server failure)	
	Built-in IR Range	Minimum 50 Mtrs Smart IR with Adaptive IR	
	Power supply (Outdoor)	PoE++ (IEEE 802.3bt) & OEM Power Adapter	
	Housing (Outdoor):	IP 66/67 or better shall be of same make as of camera	
	Surge protection	Internal/External (In case of External catalogue of surge protector shall be submitted)	
	Reset button	Shall be available;	
	Mount:	Surface/Ceiling/pipe/corner/parapet/ pendant with wall mount bracket/In Ceiling as required.	
	The camera housing and mount should be:	i. Of the same make as that of camera and suitable for the model number offered as specified by the manufacturer and should be an integrated unit.	
		ii. Should be compact and indoor/ outdoor type as required.	
		iii. Should support the weight of camera and accessories such as housing, pan & tilt head in any vertical or horizontal position etc.	
	Camera License	Camera should be supplied with a valid, perpetual license for all included features	
4	Video Management Servers(VMS)		
	i). Primary VMS		
	Application Module for Primary VMS	VIDEO RECORDING SOFTWARE FOR 100 CAMERAS (Including Licenses for 100 Cameras)	
	Server Configuration for primary VMS	CPU: 64-BIT HIGH PERFORMANCE, MINIMUM TWO PROCESSOR OF 16 CORE EACH OR HIGHER INTEL/AMD CPU OPERATING AT 3.2 GHZ OR MORE WITH A MINIMUM OF 60 MB CACHE OR HIGHER	

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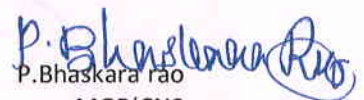
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	Graphic Card	4Gb or Higher	
	RAM (GB) Per server	64 or higher	
	Internal Storage	2X 480GB SSD RAID1 FOR OS & APPLICATION	
	OS	Windows Server 2019/2022 or Linux-based Server	
	NIC Card	2X1 GBE AND 2X10 GBE NETWORK INTERFACE	
	Chassis	Rack mountable	
	ii) Failover VMS		
	Application Module for Primary VMS	VIDEO RECORDING SOFTWARE FOR 100 CAMERAS (Including Licenses for 100 Cameras)	
	Server Configuration for Failover/Redundant VMS	CPU: 64-BIT HIGH PERFORMANCE, MINIMUM TWO PROCESSOR OF 16 CORE EACH OR HIGHER INTEL/AMD CPU OPERATING AT 3.2 GHZ OR MORE WITH A MINIMUM OF 60 MB CACHE OR HIGHER	
	Graphic Card	4Gb or Higher	
	RAM (GB) Per server	64	
	Internal Storage	2X 480GB SSD RAID1 FOR OS & APPLICATION	
	OS	Windows Server 2019/2022 or Linux-based Server	
	NIC Card	2X1 GBE AND 2X10 GBE NETWORK INTERFACE	
	Chassis	Rack mountable	
	iii) Video Management Software Specifications		
	Scalability & Compatibility	Shall Support unlimited IP cameras, scalable via additional licenses and hardware. Must be open-platform, brand-agnostic, and ONVIF-compliant (Profiles S, G, T, M)	
	Client-Server Architecture	Centralized management of video, audio, and data across an IP network, supporting multi-site and multi-server installations	
	Licensing & Platform	Windows-based perpetual license. Supports mobile (iOS, Android) and web clients	
	Automation & Failover	Intelligent device detection, auto-configuration, and native failover for management and recording servers (switch-over within 30 seconds).	
	Recording & Storage	Supports H.265+, H.265, H.264, MPEG4, and MJPEG with multi-format export. Storage options include NAS, iSCSI, DAS, and local/network drives.	
	Security & Cyber Compliance	Must have ISO certifications, undergo security audits as per OWASP, and support encrypted streaming (RTSPS/SRTP/RTSP over TLS).	
	Monitoring &	Supports live view for 100+ cameras, digital zoom,	


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	Playback	event-based search, and intelligent video analytics.	
	Remote & Mobile Access	Live video streaming, PTZ control, and alerts via mobile apps and web clients.	
	Administration & User Roles	Centralized management with role-based access, automatic health checks, and log generation	
	Multi-Format Streaming	Supports RTSP, HTTP, RTP over TCP/UDP for optimized network bandwidth utilization	
	Automated Event Notifications	Configurable alerts via email, SMS, phone call, or pop-up notifications.	
	Edge Recording Synchronization	Auto-syncs camera recordings in case of network failure via ONVIF G Profile.	
	Multi-Layered Backup	Automatic archiving, failover storage, and redundant storage management.	
	Customizable Video Wall	Supports multiple monitors, configurable grid layouts, and adaptive streaming	
	Multi-User Collaboration	Shall Allow simultaneous access from 50+ users with role-based permissions.	
	Health Monitoring & Reporting	Real-time dashboard for camera uptime, server health, and storage usage.	
	Seamless Third-Party Integration	Compatible with PIDS, radars, IoT sensors, and access control systems.	
5	Suitable rack	Suitable for housing • Dual servers, • NAS Storage, • Dual 24 port switch , • KVM switch, • LIU, Patch panels, & • Additional Dual NTP Servers provided by AAI	
6	Software Licenses		
	Client Software Licenses	Client Software for Minimum 15 Client locations	
7	NAS Storage System with RAID 6 configuration		
	NAS Storage System	NAS Storage System with RAID 6 configuration shall be used to record video streams based on the configuration assigned by administrator. Workstations & Servers within the LAN should be able to access the recorded video streams. The Storage device shall Support simultaneous play back and recording at full duplex operation.	
	Usable Storage	160 TB expandable up to 200 TB	
	Compatible	It shall provide a high-quality recording storage and playback of images. It shall support integration with	

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		LAN to provide Centralized Management and shall operate on Windows/Linux/Unix/OEM OS. Support of user management for security level control and authentication required.	
	Controller	Dual Active-Active Controllers in failover mode with no single point of failure	
	Onboard Cache	16 GB per Controller (Mirrored) or more	
	Storage Disk	10K/ 7.2K rpm SAS/NL-SAS/SATA disk for 35 days	
	HDD type support	SAS/ SSD/ NL-SAS/ SATA or compatible	
	Inbuilt RAID Support	RAID-Double Disk failure support/ RAID-6	
	Network Transport Protocols	TCP/IP	
	Protocol	Server/ Storage should support Protocol (As per solution (Any One): CIFS, NFS, iSCSI, FCP, HTTP/HTTPS, FTP, NTP, SNMP, SMTP, DHCP and DNS.	
	Power Supply	Hot pluggable Redundant Power Supply	
	Chassis	Rack mountable	
8	8 Port USB KVM Switch integrated with foldable 17" (inch) display		
	Computer Connections	• Direct - 8 or better • 256 (via Daisy-chain) or better	
	Port Selection	OSD, Hotkeys, Pushbuttons or better	
	Connectors	External Console Port :1 x SPHD-18 Male (Yellow) External Mouse Port : 1 x USB Type-A Female (White) KVM Ports : 8 x SPHD-15/17 Female (Yellow) Daisy-chain Port : 1 x DB-25 Male (Black) FW Upgrade :1 x RJ-11 Female (Black) USB Port : 1 x USB Type-A (Female) Reset : 1 x Semi-recessed Pushbutton LCD Adjust : 4 x Pushbutton LCD On/Off : 1 x LED Pushbutton (Orange) Port Selection: 2 x Pushbutton Station Selection :2 x Pushbutton	
	Emulation	Keyboard / Mouse: USB	
	Video	1280 x 1024 @ 75 Hz; DDC2B or better	
	Scan Interval (OSD Select)	1-255 sec. or better	
	Display	17inch LCD or Better	
	Operating Temp.	0-50° C	
9	High-End Workstation Computer (without Monitor)		
	CPU	• 13th generation core i7/AMD Ryzen 7 7800X3D processor or higher • operating at 3.4Ghz or more with 8MB Cache or higher	

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	Memory	64GB DDR4 or higher	
	Hard Disk	Total 1TB (500GB SSD and 500GB SATA)	
	Video card	16GB Nvidia Ge Force GTX series or better Dual port Graphics card with HDMI/Digital compatibility	
	Mouse and Keyboard	Wireless mouse and keyboard mouse pad with wrist support	
	Network adapter (NIC)	Two (02) number of GE Ports (10/100/1000 Mbps)	
	Wireless	Minimum 1 Gbps Wi-Fi Wireless Network Adapter for Desktop PC with 2.4GHz/5GHz High Gain Dual Band 5dBi Antenna Wi-Fi with Bluetooth connectivity of 4.0 or better, supports latest Windows OS version provided	
	Sound card	In-Built	
	DVD writer	DVD RW/ Blu-Ray combo internal/external	
	Ports:	2 nos. USB 3.0 at front panel and 02 nos. USB 3.0, at rear side, Ethernet LAN, 02 nos. HDMI	
	Operating System	Latest Windows OS	
	Anti-virus & Other Software	- Compatible with OS with update subscription till warranty period and CAMC - Client VMS application software including Supply of Licensed OS	
10	55" Wall/Table mountable Display		
	Screen Size	55" (inch)	
	Monitor type	- LED Signage - STQ / Professional suitable for 24*7 operation	
	Back light	LED	
	Life Time	50,000 hrs or more	
	Aspect Ratio	16*9	
	Resolution	Full HD, 1920*1080 pixel or better	
	Brightness	450 cd/m2 or better	
	Contrast Ratio (Native)	2400:1 without IPS/ 1300:1 along with IPS (In-Plane Switching)	
	Viewing angle:	178°	
	System	Supports PAL, NTSC, and SECAM	
	Operating Voltage:	Suitable for single phase AC supply 180-230 V, 50 Hz	
	Video Input Ports:	Digital Ports: HDMI (2 or more) / DVI-D; /USB Port.	
11	Workstation Computer with Monitor		
	CPU	13th generation core i7/AMD Ryzen 7 7800X3D processor or higher - operating at 3.4Ghz or more with 8MB Cache or higher	

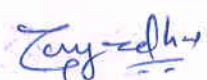
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	Memory	16GB DDR4 or higher	
	Hard Disk	Total 1TB (500GB SSD and 500GB SATA)	
	Video card	8GB NVidia Ge Force GTX series or better Dual port Graphics card with HDMI/Digital compatibility	
	Mouse and Keyboard	Wireless mouse and keyboard mouse pad with wrist support	
	Network adapter (NIC)	Two (02) number of GE Ports (10/100/1000 Mbps)	
	Wireless	Minimum 1 Gbps Wi-Fi Wireless Network Adapter for Desktop PC with 2.4GHz/5GHz High Gain Dual Band 5dBi Antenna Wi-Fi with Bluetooth connectivity of 4.0 or better, supports latest Windows OS version provided	
	Sound card	In-Built	
	DVD writer	DVD RW/ Blu-Ray combo internal/external	
	Ports:	2 nos. USB 3.0 at front panel and 02 nos. USB 3.0, at rear side, Ethernet LAN, 02 nos. HDMI	
	Operating System	Latest Windows OS	
	Monitor	• 24 Inch Full HD IPS LED Display, • Aspect Ratio: 16:9, • Resolution: Full HD, 1920 X 1080 pixels or better, • Contrast Ratio: 1000:1, • Viewing angle: min 178°, • Operating Voltage: Suitable for single phase AC supply 180-270 V, 50 Hz, • Video Input Ports: 2 HDMI, DP, VGA, USB with HDMI cables with appropriate length as per site requirements	
	Anti-virus & Other Software	• Compatible with OS with update subscription till warranty period and CAMC • Client VMS application software including Supply of Licensed OS	
		•	
12	Layer-2 (L2) Manageable , 24 Port PoE /PoE+ Network switch with RPS (Redundant power supply) (POE Power Budget 350W) or better with suitable racks		
	Physical Specifications	• Ethernet Ports: 24 No 's 10/100/1000 Base-T auto-sensing PoE+ ports • SFP Ports with modules (1G SFP Transceiver): 4 No's of dedicated 100/1000 Base-X Fiber SFP ports.	
	Performance Specification	• PoE budget: 350W or more • Bandwidth: 56 Gbps non-blocking or above	


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		- Forwarding rate: 41 mpps or more - MAC Address database size: 16 media access control (MAC) addresses or more - VLAN: 256 or above - Packet buffer: 1.5Mb or above	
	IEEE Network Protocols	IEEE 802.3 of (PoE), IEEE 802.3 at (PoE+), IEEE 802.1X Radius network access control	
	Network Security and Traffic	RADIUS based VLAN assignment via .1x MAC-based .1x Network Storm Protection Broadcast, Unicast, Multicast Protection Access Control Lists (ACLs) L2 / L3 / L4 IP ,MAC, TCP/UDP -based ACL MAC lockdown IEEE 802.1x Radius Port Access Authentication	
	L2 Services	IEEE 802.1Q VLAN Tagging Should support STP, MSTP, RSTP Broadcast Storm Control , LLDP-MED IGMP Snooping (v1, v2 and v3) , IGMP Snooping queries	
	Network Monitoring and Discovery Services	802.1ab LLDP, SNMP V1, V2, V3	
	Quality of Service(QoS)	- Advanced per-port PoE controls (enable/disable/power limit) - Diff ServQoS, IEEE 802.1p COS, WRR (Weighted Round Robin)	
	Management	- Configurable Management VLAN - switch should support both Standalone & cloud management - Dual software (firmware) image - Web-based graphical user interface (GUI)	
	Environmental Conditions	Operating Temperature: 32° to 122°F (0° to 50°C)	
	Certifications	CE mark, EN 55024 ,CCCUL listed (UL 1950)/cUL IEC 950/EN 60950, RoHS	
	Rack	Suitable for mounting 24-Switch, 24 port LIU, Patch panel & Surge protection patch panel	
13	24 port LIU , Fiber Patch cords & other accessories		
	Type	SC Simplex type	
	Mounting Type	Rack Mount / DIN Rail mounting	
	Material	ABS	
	Adapter Plate Material	Cold rolled steel	
	Number of Ports	24	
	Size	Suitable for Server Rack	

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	Operating Temperature	-40 °C to +80 °C	
	Fibre Bending Radius	≥40 mm	
	Clamps	Available on the inner surface	
	Splice Tray	Yes	
	Standard Compliance	TIA/EIA 568.C, ISO/IEC 11801, EN50173, IEC60304, IEC61754	
	SC Single mode Fibre Adaptor	As per the site requirement	
	SC Single mode 1.5m Fibre Pigtail	As per the site requirement	
	SC LC Fiber Patch cord	As per the site requirement	
14	24 port CAT-6 Patch Panels		
	Ports	24	
	Type	Unshielded (UTP)	
	Mounting Type	Suitable for Server Rack	
	Material	Cold-rolled steel or high-impact plastic with a powder-coated finish	
	Color	Black or Gray (standard)	
	Category	CAT-6	
	Cable Type	Ethernet cables (UTP, FTP, or STP)	
	Frequency Support	Up to 250 MHz	
	Data Rate	10 Gbps (Cat6a & above)	
	Insertion Force	900g max	
	Operating Temperature	-10°C to 60°C	
15	24 Port Gigabit Rack mount PoE Surge Arrestor for the 24 port L2 Switches		
	Device Type	Rack-Mountable PoE Surge Protector	
	Number of Ports	24	
	Network Compatibility	Gigabit Ethernet (10/100/1000 Mbps)	
	Mounting Type	Suitable for Server Rack	
	Nominal Voltage	48V to 57V DC (PoE standard)	
	Peak Surge Current	10 kA (per line)	
	Current Leakage	< 1 µA	
	ESD Protection	IEC 61000-4-2, Level 4 (±15kV air, ±8kV contact)	
	Lightning Protection	IEC 61000-4-5, 10 kA (8/20 µs pulse)	
	Certifications	UL, CE, FCC, RoHS compliant	
16	06 Core SM Armored OFC Cable		
		6 core OS2 (9/125 micron) Single mode, Loose Tube	

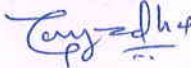
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Type	water blocking jelly, two steel wire as strengthening component, Steel tape armouring, outdoor optical Fiber cable	
Outer Jacket	PE (Polyethylene)	
Strength Members,	Two Steel wire	
Armouring	Corrugated Steel tape armouring	
Moisture Protection	Waterproof Tape	
Loose Tube construction	Water blocking Jelly	
Loose Tube Diameter	2.1mm	
Cable Outer Diameter	8.3mm	
Tensile Load	2000 N as per IEC 60794-1-2-E1	
Crush Load	2000 N/100 mm as per IEC 60794-1-2-E3	
Colour Code	ANSI/TIA/EIA-598-B	
Max attenuation	≤ 0.34 dB per km@1310 nm, ≤ 0.24 dB per km@1550nm	
Operating temperature	- 20°C to 60°C	
Weight	90 kg/km (approx.).	
Standard Compliance	ITU-T G.652.D Fiber, ISO/IEC 11801, ISO/IEC24702, ANSIANSI/TIA/EIA 568C.3, IEEE 802.3z Gigabit Ethernet, ROHS compliant Directive 2002/95/EC;	
Certification	CE	
Type	6 core OS2 (9/125 micron) Single mode, Loose Tube water blocking jelly, two steel wire as strengthening component, Steel tape armouring, outdoor optical Fiber cable	
Outer Jacket	PE (Polyethylene)	
Strength Members,	Two Steel wire	
Armouring	Corrugated Steel tape armouring	
Moisture Protection	Waterproof Tape	
Loose Tube construction	Water blocking Jelly	
Loose Tube Diameter	2.1mm	
Cable Outer Diameter	8.3mm	
Tensile Load	2000 N as per IEC 60794-1-2-E1	
Crush Load	2000 N/100 mm as per IEC 60794-1-2-E3	
Colour Code	ANSI/TIA/EIA-598-B	


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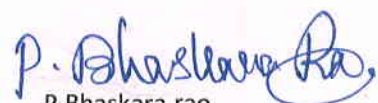
17	Cat-6 cable	
	Category	CAT -6UTP cable
	Length	305 meter (1 Bundles)
	Standards	• Complying ANSI/TIA-568-C.2 category6 ISO 11801 Class E • RoHS compliant • Only popular makes are accepted
18	1 inch PVC Pipes	
	Type	Pipes (As per site requirement)
	Size	1" (1 inch)
	Material	PVC
19	10 Feet GI Pipe poles with suitable welding structure at base	
	Material	Galvanized Iron (GI) Steel
	Coating	Hot-dip galvanized for corrosion resistance
	Outer Diameter	50mm or Higher
	Height	10ft or above
	Base Plate	300mm x 300mm x 6mm thick MS plate or better
	Camera Mounting Arm	Bracket or pole clamp for PTZ or fixed cameras



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